

MuDiS: An Audio-independent, Wide-angle, and Leak-free Multi-directional Speaker

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ABSTRACT

This paper introduces a novel multi-directional speaker, named MuDiS, which utilizes a parametric array to generate highly focused sound beams in multiple directions. The system capitalizes on air nonlinearity to reproduce sound from ultrasounds, successfully overcoming challenges inherent in traditional parametric arrays, such as transducer size and wavefront shape. It supports three important features simultaneously: independent beams, wide-angle digital steering, and unintended leakage suppression. To address these challenges, we designed a specialized cell structure that connects ultrasonic transducers, redirecting an approximately omnidirectional wavefront with optimal interspacing. An optimization-based algorithm is developed to minimize unintended leakages, and a nonlinear distortion reduction scheme is proposed to enhance sound quality. The paper showcases a prototype demonstrating the system's capabilities as a multi-directional speaker with a wide sound projection angle. Experimental results validate the effectiveness of our approach. The proposed multi-beam projection system rivals the performance of commercially available single-beam projection directional speakers, and improved steering angle and sound fidelity compared to multi-beamforming performance using traditional parametric arrays.

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CCS CONCEPTS

• **Human-centered computing** → **Sound-based input / output.**

KEYWORDS

Multi-directional speaker; Air Nonlinearity

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1 INTRODUCTION

Motivation: The ability to manipulate acoustic fields in spatial distribution holds immense potential across numerous applications. A typical application is a multi-directional speaker designed to deliver specific audio to distinct individuals separately. Consider a scenario utilizing a multi-directional speaker (depicted in Fig. 1) *In a museum, where each visitor is engrossed in a different exhibit, necessitating specific insights about the piece before them. The multi-directional speaker dispenses unique corresponding audio exclusively to each visitor without disturbing others, which allows every individual to relish a personalized experience without any external devices (e.g., earphones)* Such multi-directional speakers hold the potential to establish discrete and personalized sound realms within communal areas, including museums, gyms, libraries, and beyond.

Current multi-directional speakers primarily rely on speaker arrays. The beam direction can be controlled by altering the sound phase emitted from each speaker. However, the physical size of a speaker is proportional to the maximum wavelength of the produced sound waves. Consequently, speaker arrays often require large speakers. Furthermore, for effective beamforming, the element spacing typically remains around 10cm, contributing to an enlarged system size. Moreover, their sound fields vary across frequencies.

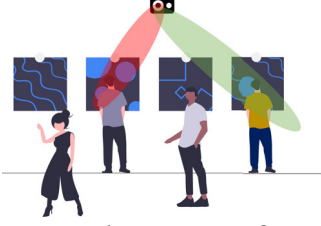


Figure 1: An example scenario for using a multi-directional speaker: in a museum, multi-directional speaker delivers audio specific to the exhibit that each visitor is viewing without interfering with each other.

Commercially available multi-beamforming speaker arrays typically support 2 to 4 beams and come in sizes ranging from 0.9m to 4.5m [4–7].

In addition to speaker arrays, parametric arrays consisting of ultrasonic transducers can produce highly directional sound beams using nonlinear acoustics. Unlike conventional speakers that generate sound through diaphragm vibration at a consistent frequency, parametric arrays create audible difference frequency waves through two or more high-frequency ultrasonic waves in a nonlinear medium (e.g., air). Notably, the difference-frequency wave maintains the same directional focus as the high-frequency waves. Leading companies in this domain, such as Holosonics [30] and Focusonics [23], utilize parametric arrays to achieve directional speakers. However, it is worth noting that these products mechanically steer the beams (e.g., employing servo motors) and are limited in supporting multiple beams. Several research also explore the feasibility in parametric-based multi-directional speakers. However, they usually suffer from grating lobes and limited steering angles ($\pm 40^\circ$) [54]. [63] supports digital steering but all beams play identical audio.

Challenges: Parametric arrays offer the opportunity to construct lightweight directional speakers. However, achieving an effective multi-directional speaker necessitates fulfilling several criteria. **a) Independent beams:** diverse audio content must be concurrently delivered in distinct directions to cater to users' preferences. **b) Wide-angle steering:** the beam must be dynamically adjustable to encompass a broad coverage. **c) Leakage suppression:** the sound should exclusively appear in the desired direction, preventing sound leakage in unintended directions. In practice, developing a parametric array capable of facilitating multi-beamforming remains a challenging task due to several factors:

Spacing between ultrasonic transducers: Commercially available ultrasonic transducers often possess a relatively large diameter, surpassing half the wavelength of their emitted frequency. This design aims to amplify energy conversion and endure elevated voltage levels, which can lead to the formation of grating lobes, causing the sound to be perceptible in directions beyond the intended target.

High directivity of ultrasonic transducers: Ultrasonic transducers commonly emit focused waves, leading to a narrower steering angle. This focus contributes to a reduced overlap between sound waves from adjacent transducers. It poses a challenge when endeavoring to steer the beam across broader angles.

Nonlinear propagation effects: Multi-beamforming introduces intricate nonlinear interactions. Carrier waves from one beam can combine with sidelobes from another, leading to leakage within the audible band (see Sec. 3.2.1). These nonlinear effects induce leaks and distort the reproduced sound, thereby affecting the sound quality.

Our approach: In this paper, we develop MuDiS, a multi-directional speaker utilizing a parametric array. We propose the following approaches to address the above limitations:

Empirical metasurface design: Inspired by recent advances in acoustic metasurfaces, we designed a unique cell structure adept at redirecting acoustic signals toward a desired wavefront (Sec 3.1). Our intricate design accounts for diverse physical interactions challenging holistic shape optimization. We decomposed and iteratively optimized it, minimizing energy loss and ensuring a compact exit size (i.e., equal to half-wavelength) for practical implementation.

Optimization-based beamforming: Diverging from multi-beamforming systems employed in wireless communication [9, 22, 69] and sensing [57, 87], our objective transcends mere signal enhancement toward the intended direction. We strive to preclude sound leakages to undesired directions. However, nonlinear acoustics introduces additional leakage due to interactions among carrier waves from different beams. To tackle this, we propose an optimized beamforming approach to curtail these unintended leaks. Furthermore, in practical implementation, we address phase variations over time arising from hardware defects (Sec. 3.2).

Nonlinear distortion reduction: In addition to the leakages, the audio remains susceptible to distortion arising from nonlinear interactions. To enhance sound quality, we introduce a nonlinear distortion reduction scheme that formulates an optimization problem to minimize the disparity between down-converted audio and the intended audio (Sec. 3.3). This approach can elevate perceived sound quality and overall user experience.

Extensive experiments verify the efficacy of MuDiS. It successfully suppresses unintended leakages, surpassing the multi-beamforming capabilities of traditional parametric arrays. Moreover, MuDiS handles a broader steering angle range, from 50° to 130° half-power beam width (HPBW) via our metasurface. The results underscore the extensive applicability of MuDiS as a multi-directional speaker (refer to supplementary video ¹).

¹Supplementary video: <https://acousticmudis.github.io>

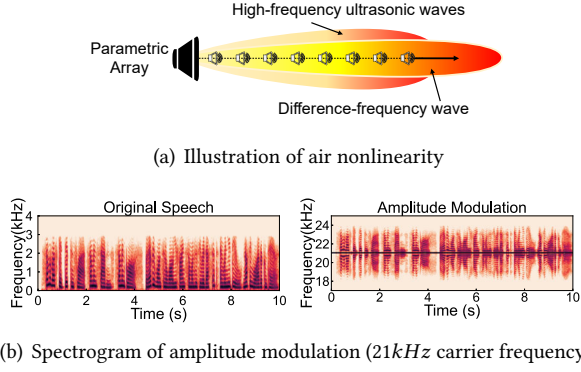


Figure 2: The illustration of air nonlinearity and an example of modulating audible sounds to a parametric array. The carrier frequency is 21kHz and the low frequency (e.g., 1Hz–3kHz) is up-converted.

Our contributions can be summarized as follows:

- We implemented a multi-directional speaker based on a parametric array, facilitating broad angles for delivering diverse audio content along various directions.
- We designed a unique cell structure to overcome the physical properties of ultrasonic transducers and form a parametric array with optimal interspacing and an approximate omnidirectional wavefront. Our study encompassed the impact of shape and material, and we furnished simulation guidelines.
- We proposed an optimization-based beamforming scheme to suppress unintended leakages in multi-beamforming in practice.
- We improved the perceived sound quality and user experience by mitigating nonlinear distortion.
- We built a prototype of MuDiS. Experiment results show MuDiS can perform better than multi-beamforming using traditional parametric arrays in terms of sound quality and steering angle.

2 BACKGROUND

2.1 Parametric Array

Parametric array [56] is a nonlinear acoustic transduction mechanism that can generate audible frequency difference waves through the nonlinear interaction of two or more ultrasonic waves in the air (shown as Fig. 2(a)), which is called **air nonlinearity**. Air nonlinearity occurs in the propagation of sufficient large sound waves, which exhibits nonlinear properties, as theoretically expressed as KZK equation [20]. Simply express this equation, the received signal $r(t)$ for a transmitted signal $s(t)$ can be derived as:

$$r(t) = A_1 s(t) + A_2 s^2(t) + A_3 s^3(t) + \dots \quad (1)$$

where A_1, A_2, A_3 are the attenuation coefficients for the first-, second-, and third-order nonlinear terms.

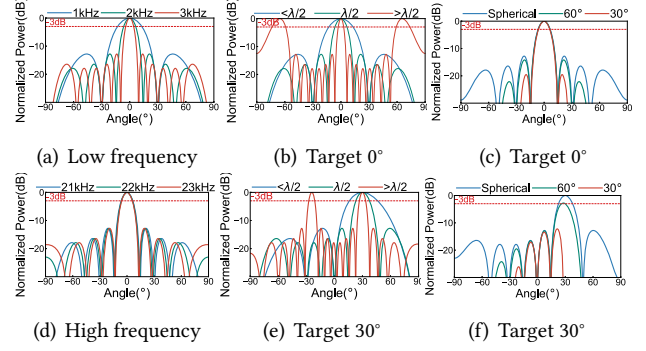


Figure 3: Simulated beam patterns of an 8-channel phased array: 3(a) and 3(d) represent beam patterns at varying frequencies, where the element spacing is half-wavelength. 3(b) and 3(e) depict beam patterns with different element spacing ($< \lambda/2$, $\lambda/2$, and $> \lambda/2$). 3(c) and 3(f) showcase beam patterns with distinct wavefront shapes (spherical, 60° HPBW, 30° HPBW).

In practice, the square (second-order) term offers the opportunity to reproduce sound from ultrasound, while the higher-order terms are fragile and negligible in practice. One straightforward modulation scheme is amplitude modulation [86]. Suppose a low-frequency audio signal $h(t)$, and a carrier with central frequency f_c , the modulated signal is $s(t) = h(t) \cos(2\pi f_c t) + \cos(2\pi f_c t)$. Fig. 2(b) shows an example of amplitude modulation. Since human hearing is not sensitive to high-frequency sounds, it can be regarded as a low-pass filter. Then we apply it to the square term of Eq. 1, and the low-frequency signal can be derived:

$$s_{low}(t) = \frac{A_2}{2} (2h(t) + h^2(t) + 1) \quad (2)$$

This approach allows for the reproduction of the desired audio $h(t)$ using ultrasound. Thanks to the ultrasonic transducer's high directivity, the parametric array serves as a directional speaker [23, 30]. In contrast to conventional speaker arrays [65], a parametric array markedly reduces device sizes. However, existing implementations are limited to a single beam and mechanical steering (e.g., servo motor control). In our endeavor, we strive to develop a parametric array capable of supporting multiple freely steerable audible beams.

2.2 Phased Array

2.2.1 Single-beam beamforming. A phased array [28] is a system of N antennas or transducers, where each element i ($1 \leq i \leq N$) emits a wave multiplied with a complex weight which represents both amplitude and phase. The distance between adjacent elements is d . By changing the complex weight, we can change the beam pattern and steer the main lobe of the beam in any direction. Assuming the set of all possible transmission directions is Θ . Specifically, the beam

pattern towards direction ϕ can be expressed as:

$$W(\phi) = \sum_{i=1}^n w_i e^{j2\pi \frac{d}{\lambda} (i-1) \sin \phi} \quad (3)$$

where j is the imaginary unit, λ is the wavelength. The resulting wavefront can be steered by adjusting the complex weight. For instance, if our aim is to perform beamforming towards direction θ within the set Θ , the complex weights are configured as $w_i^\theta = e^{-j2\pi \frac{d}{\lambda} (i-1) \sin \theta}$. This setup enables the summation of signals from all channels along the intended direction, thereby achieving effective beamforming.

The beamforming performance of a phased array is affected by (i) **frequency**: With the same bandwidth, high-frequency beamforming generally behaves superior to low-frequency beamforming due to a more consistent beam pattern (Fig. 3(a) and 3(d)). (ii) **Element spacing arrangement**: Narrow element spacing increases the beam width while excessive spacing produces grating lobes (Fig. 3(b) and 3(e)). The best element spacing satisfies $d \leq \frac{\lambda}{2}$. (iii) **Type of wavefront**: The type of wavefront limits the gain of the main lobe at different angles of the phased array focus, and the spherical wave has a more stable gain effect at different angles compared to the focused wave (Fig. 3(c) and 3(f)).

2.2.2 Multi-beam beamforming. So far, we have established one-beam beamforming whose beam weight vector for a single beam can be expressed as $\mathbf{w}_\theta = \frac{1}{\sqrt{n}} [1, w_2^\theta, \dots, w_n^\theta]^T$. Assuming that there are M -beam needs to point to, denoted by $\theta_1, \theta_2, \dots, \theta_M$, the corresponding beam weight vectors are $\mathbf{w}_{\theta_1}, \mathbf{w}_{\theta_2}, \dots, \mathbf{w}_{\theta_M}$. We can construct a beam weight for multi-beam beamforming by adding the beam weight vectors of each target angle, expressed as $\mathbf{w} = \sum_{j=1}^M \mathbf{w}_{\theta_j}$. To ensure the total power constant, we scale the value to satisfy the constraint $\|\mathbf{w}\| = 1$. Note that each target direction will be interfered with by signals from other angles. We tried to suppress it by leveraging an optimization (Sec. 3.2).

■ **Key Idea.** Combine air nonlinearity and phased array to achieve a multi-directional speaker, while challenges remain in improving steering angle and sound quality.

3 MULTI-BEAMFORMING PARAMETRIC ARRAY

In this paper, we present MuDiS, a cost-effective and compact parametric array-based multi-directional speaker. It supports multiple concurrent beams steering over a wide range and suppresses sound leakages toward unintended directions. To achieve MuDiS, we propose the following techniques. First, we design an **acoustic metasurface** (Sec. 3.1) to redirect acoustic signals from ultrasonic transducers to ensure the desired spacing distances and preferred wavefront while

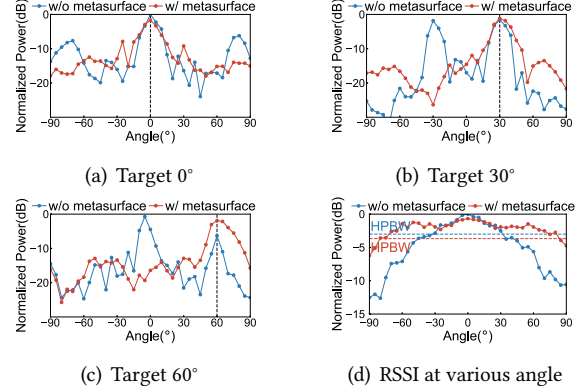


Figure 4: The measured beam pattern of the high frequency of a parametric array with and without metasurface. The metasurface can correct the wavefront shape and improve the steering angle.

maintaining a compact size and high energy output. Second, we proposed an **optimization-based beamforming scheme** (Sec. 3.2) to minimize the leakage caused by nonlinear interaction between beams. Finally, we propose an **nonlinear distortion reduction scheme** (Sec. 3.3) to confront nonlinear distortion and restore sound quality.

3.1 Empirical Acoustic Metasurface Design

3.1.1 Motivation of Metasurface Design. The target for MuDiS is to support a wide-angle dynamic beam steering without interfering with other directions. However, traditional parametric arrays (e.g., existing directional speakers) failed to satisfy the demand. This is because commercial-of-the-shelf (COTS) ultrasonic transducers have a diameter of more than half the wavelength of their emitting frequency in order to achieve higher energy conversion efficiency and withstand voltage levels [3, 33, 40, 51]. In this case, **grating lobes** will inevitably occur. Moreover, the **directional focusing beam** transmitted by ultrasonic transducers leads to a worse beam pattern, significantly limiting the freedom of beam steering.

To explain this, we conducted a preliminary experiment. Fig. 4 shows a measured beam pattern of an 8-channel parametric array emitting 21kHz waves. The diameter of each ultrasonic transducer is 16mm, which restricts the minimum element spacing. By changing beam weights, the beam is steered to target 0°, 30°, and 60°. It can be noticed that grating lobes appear at -30° and -5° while the expected direction are 30° and 60°, respectively. In such a particular case (i.e., target 60°), the energy of grating lobes (-5°) is much larger than the main lobe along the intended direction (60°) due to the combination of both grating lobes and the focusing beam. It will significantly produce leakages that reduce the performance of a parametric array to steer the beam.

In this work, we introduce an acoustic metasurface coupled with each ultrasonic transducer to address the above

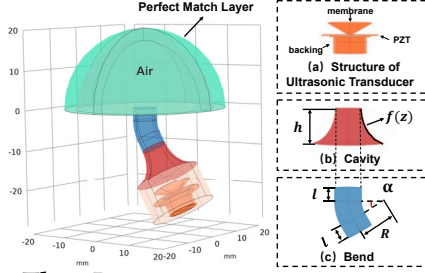


Figure 5: The 3D geometry components of physical simulation of an acoustic metasurface.

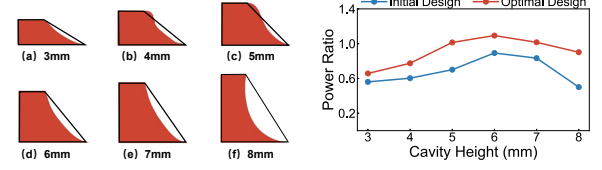
issues. The purpose is to maintain proper element spacing (i.e., equal to half-wavelength) and desired output wavefront (i.e., approximate spherical wave) while minimizing energy loss. Since the phase shift caused by the metasurface for each element is the same, it doesn't affect the pre-set beamforming weights for steering.

3.1.2 Simulation Configuration. To achieve an effective metasurface with an intended outgoing wavefront and minimize energy loss, we carefully design it through a physical simulator (e.g., COMSOL [2], which is based on finite element analysis). Specifically, our acoustic metasurface's 3D geometry comprises at least two parts: a cavity that reflects acoustic energy towards the narrow outlet, and a bend that connects this cavity. When the sound wave passes through the metasurface, the sound can get amplified or attenuated [78, 90]. In order to simulate the strength of the outgoing audio signal, the built physical simulation contains four 3D geometry components (Fig. 5): a metasurface composed of a cavity and a bend, a perfect match layer consisting of air to represent infinite air domain, and a 3D structure of an ultrasonic transducer. Several principles should be employed in the entire physical simulation process:

Multipath effect: The sound waves emitted by the ultrasonic transducer hit the wall of the metasurface and reflect many times. The different time delay for each acoustic component to reach the junction plane of the cavity and the bend result in distinct amplitude and phase, which enhance or attenuate the outgoing waves at the junction plane. The objective is to make the energy emitted as large as possible.

Wave propagation in duct: The signals coming out of the cavity end up in the bend. According to the theory of acoustic propagation in the duct [37], considering a sound wave of frequency f and speed c transmitted in a cylindrical duct of radius a , the cut-off frequency of the duct is $f_c = 1.84 \frac{c}{2\pi a}$. If $f < f_c$, the acoustic wave will propagate as a plane wave. The exit of the plane wave from the nozzle will evolve into a wavefront that approximates a spherical wave. Fortunately, the radius satisfies $a \leq \frac{\lambda}{4} = \frac{c}{4f} < \frac{1.84c}{2\pi f}$. That is to say, when the element spacing is half-wavelength, the output must be an approximate spherical wave.

Moreover, the thermoviscous loss [52] in the duct should also be considered. The impedance grows with the tube



(a) Cavity shape after optimization (b) Impact of height of the cavity.

Figure 6: The simulation result of optimization in terms of cavity height. We choose the optimized cavity of 6mm for best performance.

length. As for propagation in a bend, the bending angle should also be considered [21]. Minimizing the thermoviscous loss can boost the energy emitted. As suggested in [19], we choose smooth inner surface to avoid viscous dissipation and energy loss.

Radiation impedance: Previous acoustic metasurfaces that Owlet [25] is designed for microphones without considering the sound source, and SPiDR [11] leverages consistent pipe cross-sections with few wave reflections. However, we aims for tapered output, so the reflected waves on the sound source should be considered. The reflected sound wave from the cavity will couple with the vibration of the ultrasound transducer, thereby suppressing its vibration, so-called radiation impedance [81]. Therefore, the geometry of the transducers should be considered and their parameters are accessible through the manufacturer's datasheet. Given that the optimized shape can be irregular to be expressed analytically, the computation of the impact resulting from multiple reflections of sound waves becomes intricate. To address this, we conducted simulations involving acoustic-solid coupling within COMSOL, leveraging finite element analysis.

Summary. In our prototype, we designed a cell structure for an ultrasonic transducer with a diameter of 16mm and a central frequency of 21kHz. Since our 3D printer supports 0.2mm accuracy, and the wall thickness of the cell needs at least 1mm, the exit radius was set to 3.1mm to meet the requirement of a half-wavelength (i.e., 8.2mm). Higher frequencies demand more precise printers for finer spacing, and the metasurface will be designed accordingly. Specifically, the critical design parameters in our setup are as follows: 1) cavity height, h , varies from 3mm to 8mm in steps of 1mm, 2) cross-sectional area variation within the cavity denoted by $f(z)$, where z ranges from 0 to h , 3) bend length, l , spans from 2mm to 8mm in 1mm increments, 4) bend angle, α , ranges from 10° to 90° in 10° steps, and 5) bend inner radius, R , varies between 8mm and 13mm in 1mm intervals. A visual representation of these parameters is presented in Fig. 5.

3.1.3 Optimal metasurface design. The ideal acoustic metasurface should make its outgoing sound energy as large as possible without affecting beamforming performance. Since the metasurface on each ultrasonic unit is the same, the phase

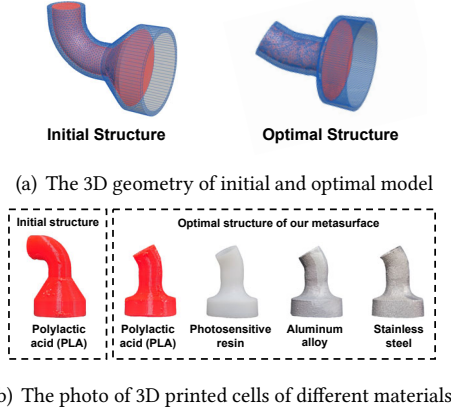


Figure 7: The 3D geometry and 3D printed structure of initial and optimal structure. Four materials are selected for comparison. The wall thickness is 1mm.

difference for each channel output will not change. Thus, the metasurface will not affect beam steering. The optimization goal is to maximize the sum of the energy of each frequency component (N frequency bins from f_1 to f_N) of points on an arc of distance s from the output plane. We define the following objective function:

$$\max_{\vec{P}_c, \vec{P}_b} \sum_{s, \theta} \sum_{f=f_1}^{f_N} p(f)$$

The optimization of the metasurface is performed using the COMSOL optimization module [2]. Specifically, the sound field generated by an ultrasonic transducer is simulated by the piezoelectric effect coupled with Solid Mechanics and Electrostatics modules. A pressure acoustic module simulates the propagation of sound waves. Thermoviscous acoustic impedance is applied to the wall to simulate the thermoviscous loss. To simplify the computational complexity and searching space, we decomposed the parameter of the metasurface into two parts: a cavity $\vec{P}_c = \{h, f(z)\}$ and a bend $\vec{P}_b = \{l, \alpha, R\}$ to maximize the objective.

Cavity optimization: The 3D geometry of the cavity can be simplified to a 2D axisymmetric model to reduce computational complexity. The initial cross-sectional shape is a trapezoid, the lower base is the radius of the transducer, the upper base is half-wavelength of the frequency, and the height is set h . The variation of the cross-sectional area along the z -axis can be transformed into a deformation of the hypotenuse of the trapezoid. The shape optimization is gradient-based, utilizes the Polynomial Boundary feature, and handles the deformation through smoothing equations. In contrast, the deformation is given by $\mathbf{d} = \sum_i^n B_i^n(s) \mathbf{c}_i$, ($-d_{max} \leq c_i \leq d_{max}$). Here, B_i^n is the i th order of a Bernstein polynomial of order n to achieve smooth deformation. d_{max} is the maximum displacement, which implies the horizontal movement of every point is within length d_{max} . Here, we set $d_{max} = h/5$. Fig. 6

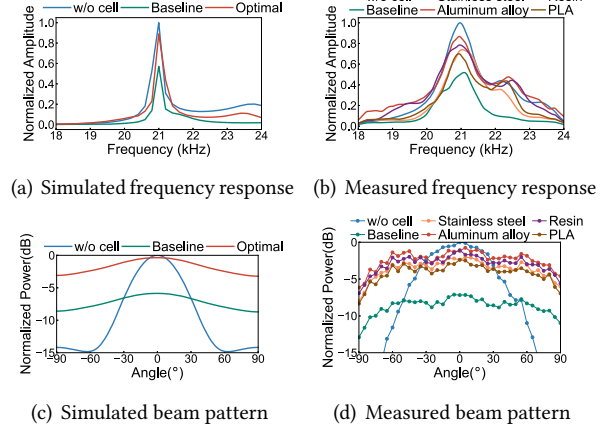


Figure 8: The simulated and measured frequency response and beam pattern of our designed cell structure.

shows the optimized height results and calculates the simulated power ratio to the power without metasurface. The results indicate that the optimized structure of 6mm is the best. Thus, we chose it as the best cavity shape.

Bend optimization: The 3D geometry of the cavity will be taken as a preset to be assembled with the bend. The bend has the same cross-section as the upper part of the cavity. The initial settings are $l = 2\text{mm}$, $R = 8\text{mm}$, and $\alpha = 90^\circ$, chosen as the baseline. The simulator takes the Nelder-Mead scheme [67] to find the optimal combination of bend length l , bend angle α , and inner radius R . Fig. 7(a) and Fig. 7(b) show the 3D geometry of the structure of the cell and the corresponding 3D printed cell of different materials. Fig. 8 compares the optimal metasurface structure with the initial design, and the result shows that the optimized structure outperforms 7dB to the baseline. Moreover, the metasurface provides wider HPBW (half-power beam width) to bring a wider steering angle.

Material selection: Apart from the basic structure, we also study the impact of materials. Four materials are selected: stainless steel, photosensitive resin, aluminum alloy, and polylactic acid (PLA), which cost \$23, \$4.5, \$42, \$1.5 per 100g, respectively. Fig. 8(b) and Fig. 8(d) show that the material minimally impacts the beam pattern. PLA and stainless steel perform worse due to the rough inner wall. In comparison, photosensitive resin and aluminum alloy are relatively smooth, so less power loss. For both performance and cost considerations, we have chosen photosensitive resin as the material we use.

Summary. We experimented with measuring beam patterns after covering an acoustic metasurface. Fig. 4 shows that our metasurface efficiently suppresses grating lobes and improves steering angle (from 60° to 170° HPBW). However, it cannot realize free leakages in other directions when projecting multi-beams. The next section will address the reasons and propose optimized algorithms to rectify the issue.

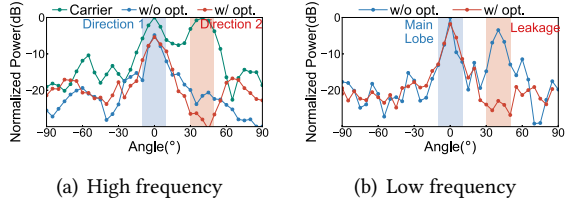


Figure 9: The comparison of measured beam pattern with and without optimization. The carrier band in two directions strengthens the side lobe in unintended directions, resulting in leakages in low frequency. MuDiS can effectively eliminate the leakage.

3.2 Optimization-based Beamforming

MuDiS is designed to project specific content in each direction. While the metasurface can effectively reduce the grating lobes, sidelobe still exists. Unfortunately, the side-lobe of one beam will mix with the carrier of the main lobe of another beam, resulting in unintended leakages.

3.2.1 Unintended Leakage production. To explain the unintended leakage production through multi-beam projection, we take 2 beams as an example without loss of generality. Suppose a phased array with n channels, the central frequency of the carrier wave is f_c , the audios to be reproduced along two directions θ_1 and θ_2 are $h_1(t)$ and $h_2(t)$, respectively. Thus, the two signals $s_1(t)$ and $s_2(t)$ modulated by amplitude modulation are $s_1(t) = h_1(t) \cos(2\pi f_c t) + \cos(2\pi f_c t)$ and $s_2(t) = h_2(t) \cos(2\pi f_c t) + \cos(2\pi f_c t)$. To steer the beams to their respective directions, the output signal in multi-beamforming for the i th channel according to Sec. 2.2.2 is $s_1(t)\mathbf{w}_{\theta_1} + s_2(t)\mathbf{w}_{\theta_2}$. In this case, the received high-frequency signal towards one of the target directions θ_2 is

$$\begin{aligned} R_{High}^{Multi}(\theta_2) &= A \sum (s_1(t)\mathbf{w}_{\theta_1} + s_2(t)\mathbf{w}_{\theta_2}) e^{j2\pi \frac{d}{\lambda_i} (n-1) \sin \theta_2} \\ &= A n (\alpha s_1(t) e^{j\sigma} + s_2(t)) \end{aligned}$$

Here, α and σ represent the side lobe's amplitude change and phase shift on signal $s_1(t)$, where $\alpha < 1$. Afterward, we apply the above equation to Eq. 1 and performs a low pass filter (LPF) to obtain audible sound heard by human at θ_2 :

$$\begin{aligned} R_{Low}^{Multi}(\theta_2) &= LPF(A_2(R_{High}^{Multi}(\theta_2)^2)) \\ &= A_2 A^2 n^2 LPF \left[\underbrace{s_2(t)^2}_{Expected} + \underbrace{2\alpha s_2(t)s_1(t)e^{j\sigma}}_{Leakage} + \underbrace{\alpha^2 (s_1(t)e^{j\sigma})^2}_{Ignored} \right] \end{aligned}$$

Here, $s_2(t)^2$ is the expected signal at θ_2 to reproduce original audio $h_2(t)$. Since α^2 is very small, so $\alpha^2 (s_1(t)e^{j\sigma})^2$ can be ignored. However, the second term will bring in a carrier wave in $s_2(t)$ resulting in a **sound leakage** of $2\alpha h_1(t)$. Similarly, there will be an $h_2(t)$ leakage in θ_1 . As for the single beam, the leakage in the side lobe is only $\alpha^2 h_1(t)$, which is too small to be noticed. From this, when realizing multi-beamforming,

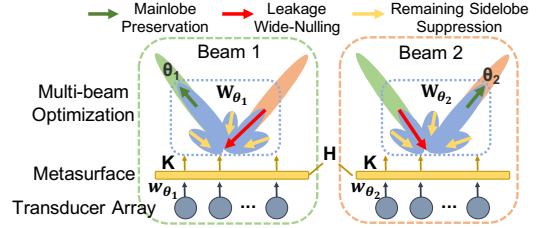


Figure 10: Problem formulation and objectives for leakage-cancellation optimization.

the carrier band of the main lobe will strengthen the side lobe of the signal of another direction, resulting in unintended leakages. Fig. 9 shows the measured beam pattern in both high and low frequency when targeting 0° and 40° . Although the side lobe in high frequency in 40° is small (blue line in Fig. 9(a)), the leakage (blue line in Fig. 9(b)) inevitably appears. This suggests that the side lobe in the direction of another main lobe needs to be further suppressed.

3.2.2 Optimization for Leakage-cancellation. This section proposes an optimization framework to eliminate the leakages as mentioned earlier.

Problem formulation: Suppose an n -channel parametric array expected to steer towards a direction set Θ , among which a beam steered in a certain direction θ , the beam weight is $\mathbf{w}_\theta$. Each transducer is connected to a cell structure designed in Sec. 3.1. The cell modifies $\mathbf{w}_\theta$ in terms of phase and amplitude, denoted by an $n \times n$ diagonal matrix H . The i -th diagonal element of H specifies how the i -th cell translates the codeword from the i -th transducer into the outgoing signal. Therefore, the outgoing signal from the metasurface S_{out} can be derived as $S_{out} = H\mathbf{w}_\theta$. The received signal W towards a receiving directions set Φ according to Eq. 3 can be derived as follow:

$$W = K S_{out} = K H \mathbf{w}_\theta \quad (4)$$

where W is an array of the same length as Φ (each element represents the received signal from a given direction ϕ), and K is a $\Phi \times n$ matrix specifying the steering vectors for the n -cell metasurface. The problem is to optimize the beam weight of each channel $\mathbf{w}_\theta$ and formulate an ideal beam pattern in terms of the power P of the received signal ($P = 10 \log_{10} |W|^2$).

Specifically, several components should be taken into account (shown as Fig. 10) during the optimization:

Mainlobe preservation: In order to maintain the strength of the mainlobe, we maximize the beam transmitted towards the direction θ . The beam width should not be so narrow that only one ear can hear the sound, which is annoying. Assuming a beam width with an interval 2α degrees wide, the main lobe direction range is $[\theta - \alpha, \theta + \alpha]$. The sum of power in the main lobe direction can be expressed as:

$$L_{main} = \sum P \times \text{mask}([\theta - \alpha, \theta + \alpha])$$

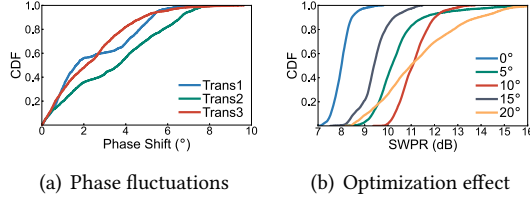


Figure 11: The CDF of phase fluctuations over time and the optimization effect for using phase robustness scheme under the impact of phase shift ranges.

where $mask(\cdot)$ is a function that returns a mask sequence of the same length as P , whose value is set to 1 within the input range and the rest set to 0. We empirically define α as 10° , as a range of 20° ensures that the human ear can hear.

Leakage wide-nulling: Besides the target direction θ , some other directions need to project audio as well. Those directions are denoted as Θ_l . As mentioned in Sec. 3.2.1, those directions emit high-frequency carriers that will become the predominant leakage directions. Our optimization is designed to suppress such leakage within a certain angle range (so-called wide-nulling) to improve the hearing experience. Assuming an interval 2β degrees wide around Θ_l , then we propose to restrain these angles:

$$L_{null} = \sum P \times mask([\Theta_l - \beta, \Theta_l + \beta])$$

where the value of β will be further discussed in Sec. 4.2.

Remaining sidelobe suppression: In addition to the target angle θ and other emission directions Θ_l , we must also suppress the remaining sidelobe to prevent possible leakages. The remaining angles are $\Theta_m = \Theta - \{\theta\} - \Theta_l$. For these angles, we take the peak value in the remaining sidelobe, and the energy can be expressed as:

$$L_{remain} = \sum peak(P \times mask(\Theta_m))$$

where $peak(\cdot)$ calculates the peak in remaining sidelobe.

■ **Objective function.** The target is to suppress leakages and side lobe energy as much as possible while maintaining the main lobe. Thus, putting the above together, we have the following objective function:

$$\begin{aligned} \max_{\mathbf{w}_\theta} \quad & L_{main} - \delta L_{null} - \eta L_{remain} \\ \text{s.t.} \quad & |\mathbf{w}_\theta| \leq 1 \quad (\theta \in \Theta) \end{aligned} \quad (5)$$

where δ and η represent the weights of nulling and suppression, we restrict the amplitude to be within 1 instead of exactly equal 1. First, since the constraint set contains the entire hypercube and includes its interior, it becomes a convex problem and is easier to solve. Second, the flexibility of choosing a more minor power allows us to minimize the sidelobes, even if sacrificing some of the main lobe power. To determine the optimal parameters, we initially conducted a multi-resolution search to identify general trends and followed it with a fine-grained search.

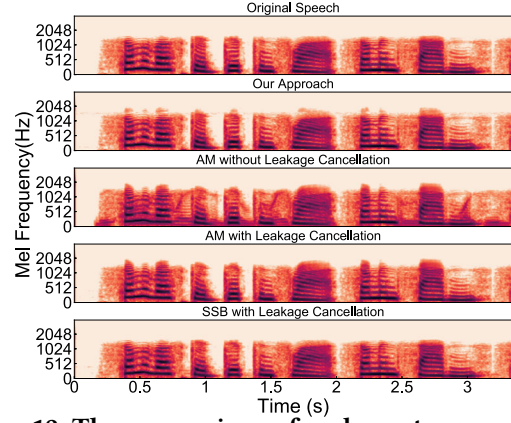


Figure 12: The comparison of mel-spectrogram under different modulation schemes. MuDiS performs the most similar mel-spectrogram to the expected audio.

Wideband beamforming: To extend the narrowband beamforming optimization to wideband signals, we first transform the input wideband audio x into the frequency domain using a Fast Fourier Transform (FFT). For each frequency, we apply the narrowband beamforming optimization to obtain the corresponding optimized codeword $\mathbf{w}_{\theta,f}$ for each transducer. Finally, we sum up the weighted signal of all frequencies and take an inverse FFT to reconstruct the acoustic signal transmitted by each transducer.

3.2.3 Optimization for Phase Robustness. Practical implementing MuDiS suffers from **hardware imperfections** [46] that the transmission phase of each transducer fluctuates when working at high power. This issue is often ignored because it does not significantly impact the main lobe, but it can greatly affect the nulling performance in Sec. 3.2.2. We recorded the phase change of 3 different ultrasonic transducers over 5 minutes and showed the results in Fig. 11(a). It indicates two phenomena: 1) Different transducers perform different phase changes; 2) The phase shift is not constant but fluctuates over time. This makes it hard to calibrate the phase of the transducer at a time.

To address the above hardware imperfections, we introduce an additional phase shift Φ_h for considering and tolerating the possible phase deviations of each channel. It considers the actual phase offset of the system to fall in the interval $\Phi_h = [-\phi_h, \phi_h]$. In order to make the optimization more in line with the actual condition, we measured the average probability density function of transducers in phase shift and assigned it as a weight function Q . Then, we introduce the compensation for the previous optimization by multiplying the phase shift; the received signal in Eq. 4 is modified as:

$$W_r = \sum KHQ\mathbf{w}_\theta e^{j\Phi_h}$$

It considers the fluctuations brought by hardware imperfections, and we apply it to Eq. 5 and find the optimal $\mathbf{w}_\theta$. In our system, we experimented to determine the value of ϕ_h . We

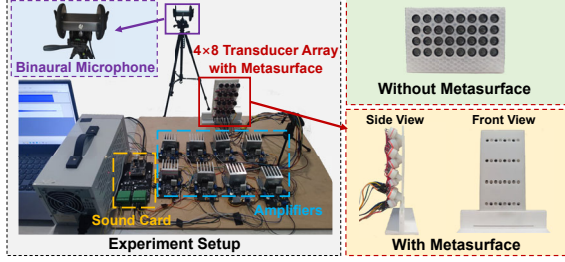


Figure 13: Experiment Setup of MuDiS.

measured the variation of SWPR (defined in Sec. 4.1) during 5 minutes, and Fig. 11(b) showed that when $\phi_h = 10^\circ$, the leakages are stably suppressed.

■ **Summary.** Our optimization practically suppresses possible leakages, solely involving the direction set and unrelated to audio content. Compared to optimizing a beam pattern for each audio separately, this significantly reduces computational complexity and storage space. Our optimization processes 1s audio with 4 beams in only 0.041s, allowing real-time audio playback with continuously dynamic beam angle adjustments for mobility use.

3.3 Nonlinear Distortion Reduction

Above, we suppress the leakages by optimization. Besides, MuDiS also suffers from **nonlinear distortion**. Note that in Eq. 2, the low-frequency components not only contain $h(t)$ that we want but also contain $h^2(t)$ term, which causes distortion. To overcome such a nonlinear distortion problem, we convert it to an optimization problem.

Assuming there are M beams to be projected with $h_j(t)$, ($1 \leq j \leq M$), the corresponding target angles are $\theta_1, \theta_2, \dots, \theta_M$. Then the traditional output signal for the phased array is $S = \sum_{j=1}^M s_j(t) \mathbf{w}_{\theta_j}$, where $\mathbf{w}_{\theta_j}$ is the optimal beam weights in Sec. 3.2. The target for MuDiS is that the low-frequency signal is received at each angle as close as possible to the expected signal at the corresponding angle. We optimized it by adding component S_e to output signal $S + S_e$ and minimized the difference to the expected signal $h_j(t)$.

$$\min_{S_e} L_d = \sum_{j=1}^M \|LPF[(S + S_e)K_{\theta_j}] - h_j(t)\|_{2i_j}^2$$

where K_{θ_j} denotes the steering vector in corresponding direction. Since the low pass filter function is derivable, we can use gradient descent to solve it.

Optimization can effectively solve the distortion problem. We compared the time series result with two existing well-known modulation schemes: the original amplitude modulation (AM) (with and without beamforming optimization) and single-sideband (SSB) modulation². In Fig. 12, we excludes

²The SSB modulation includes upper-sideband and lower-sideband modulation: $s_{ssb} = h(t) \cos(2\pi f_c t) \pm \hat{h}(t) \sin(2\pi f_c t)$, where $\hat{h}(t)$ denotes the Hilbert Transform of $h(t)$. Here, we took upper one as an example

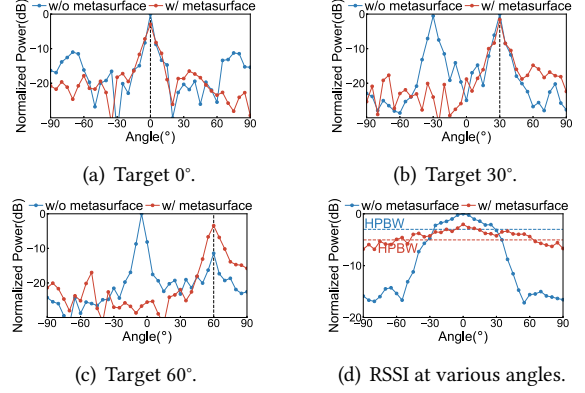


Figure 14: The beam pattern of low frequency with and without using a metasurface when targeting different angles and the RSSI at various steering angles.

the mutual leakages to compare fairly, highlighting the need for distortion reduction. The results show MuDiS that includes a distortion reduction optimization outperforms other results. Table. 1 in the evaluation also indicates the better performance of taking a distortion reduction process.

4 PERFORMANCE EVALUATION

4.1 Evaluation Methodology

Prototype: As shown in Figure 13, our prototype consists of an ultrasonic transducer array and an acoustic metasurface. The transducer array comprises 4×8 ultrasonic transducers (EU16AOF21H12T) connected to an 8-channel audio source (i.e., a sound card connected to a laptop). Each channel contains 4 ultrasonic transducers and is connected with a class D amplifier OPA541 [34], which supports a maximum power of 50W. The central frequency of the ultrasonic transducer is 21kHz. Every ultrasonic transducer is sheathed by a cell of metasurface, which is designed in Sec. 3.1.3. The distance between the output of cells of adjacent channels is 8.2mm, which is a half wavelength of 21kHz sound to eliminate grating lobes. The system is measuring $8cm(W) \times 15cm(H)$ and the total cost is \$350. For the receiver, We use a binaural microphone (Binal 2 [29]) with a 192kHz sampling rate to avoid aliasing issues. We compute the average of the sounds from two channels and take a low-pass filter with a 16kHz cut-off frequency, representing the sounds a listener would hear. We aligned the received audio with reference audio by using dynamic time warping (DTW) [50].

Testing sounds: We selected 50 English sentences generated from Text-to-Speech (TTS) generator (30 pieces) and vocal recording clips (20 pieces). Each sentence has between 10 to 50 words and includes female, male, and children voices. Each sentence was recorded for 5 times.

Performance metrics: We quantify the performance of MuDiS using 4 metrics listed below.

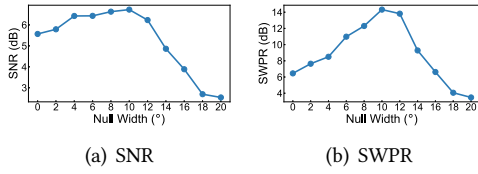


Figure 15: Impact of null width.

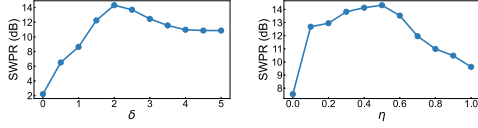
(a) SWPR under impact of δ (b) SWPR under impact of η

Figure 16: Impact of optimization weights.

1) *Signal-to-Noise Ratio (SNR)*: We use SNR to evaluate the speech recovery quality, representing the distortion to the desired signal. The SNR between a reference clean signal $x \in R^{1 \times T}$ and the received audible signal $\hat{x} \in R^{1 \times T}$ as $:SNR(x, \hat{x}) = 10 \log_{10}(\frac{\|x\|_2^2}{\|x - \hat{x}\|_2^2})$. Specifically, we use the scale-invariant SNR to reduce the impact of scale.

2) *Perceptual evaluation of speech quality (PESQ)*: PESQ is a standardized objective method [1] for assessing speech quality and generates a score ranging from -0.5 to 4.5 . PESQ is based on a psychoacoustic model that simulates the human auditory system's response to audio signals. A higher score indicates better speech quality.

3) *Mel-cepstral distance (MCD)*: MCD [41] measures the similarity between sounds accounting for the ear's critical bandwidth as a function of frequency. The details for calculating MCD can be referred to [41]. A lower MCD value means higher speech quality that human can harder distinguish.

4) *Signal-to-Worst-Point Ratio (SWPR)*: In an audible sound multi-beamforming system, grating lobes can lead to leakages to other beams or nearby individuals, diminishing user satisfaction. Here, we introduce SWPR to assess the maximum interference in other directions apart from the intended one. It is defined as the difference between the peak of the main lobe and the peak in other directions. A higher SWPR indicates reduced leakage, signifying superior performance.

4.2 Micro Benchmark

In this section, we evaluate the impact of system parameters. Two single-tone frequencies, 500Hz and 1.5kHz, are modulated and delivered in two directions.

Metasurface effectiveness. MuDiS uses acoustic metasurface to eliminate the grating lobes and improve the directional sensitivity. Fig. 14 (a)-(c) depict the low-frequency beam pattern of a 8-channel parametric array with and without metasurface when targeting 0° , 30° , and 60° , respectively.

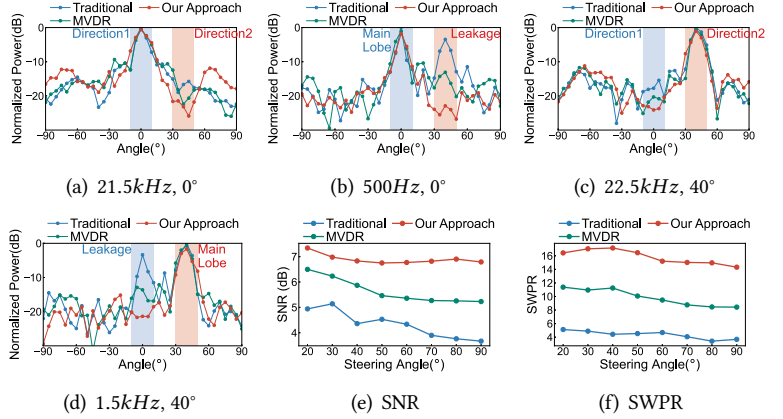


Figure 17: (a)-(d): An example of the beam pattern when utilizing different schemes. (e)-(f): The SNR and SWPR of different schemes.

Fig. 14 (d) shows the RSSI at various angles when targeting the corresponding angle. The results demonstrate that the metasurface can provide two benefits. First, it mitigates sound leakage at non-desired directions (shown as Fig. 14(b)). Second, it leads to a greater steering angle from 50° HPBW to 130° HPBW in low frequency. When targeting at an angle larger than 60° , the parametric array with the metasurface is 9dB louder than without it.

Impact of null width. MuDiS performs a wide-angle nulling optimization to suppress the leakage in unintended directions. There is a trade-off between the null width and the system performance. We run experiments where the null widths range from 0° to $\pm 20^\circ$. Fig. 15 shows the SNR and SWPR under impact of null width. Narrow null width shows poor SNR (5.5dB) due to the space occupied between the ears, while wide null width shows worse SWPR because the performance decreased in the nulling region. We select $\pm 10^\circ$ as the null width according to the results.

Impact of optimization weights. We also study the impact of weights δ and η mentioned in Eq. 5. We did a grid search to determine the parameters as $\delta \in [0, 5]$, step:0.5, and $\eta \in [0, 1]$, step:0.1. Fig. 16 shows that as δ increases, the SWPR first increases, and when it is set to 2, the SWPR drops. The SWPR decreases when η is larger than 0.5. Thus, we selected $\delta = 2$ and $\eta = 0.5$ for optimal weights in our optimization.

Impact of different schemes. We compared our optimization with two baselines: traditional beamforming and MVDR [17]. Specifically, 500Hz is always delivered to 0° , and 1.5kHz to other angles ($20^\circ - 90^\circ$). Fig. 17 (a)-(d) shows the example of the beam pattern of both high frequency and low frequency of using different schemes when delivering 1.5kHz to 40° . From the results, MVDR is not suitable for human listeners, for it only suppresses a narrow-angle point. In comparison, MuDiS can effectively suppress the leakages for achieving around 7dB SNR and 15dB SWPR in whole steering angles, which outperforms traditional beamforming and MVDR (Fig. 17(e) and Fig. 17(f)).

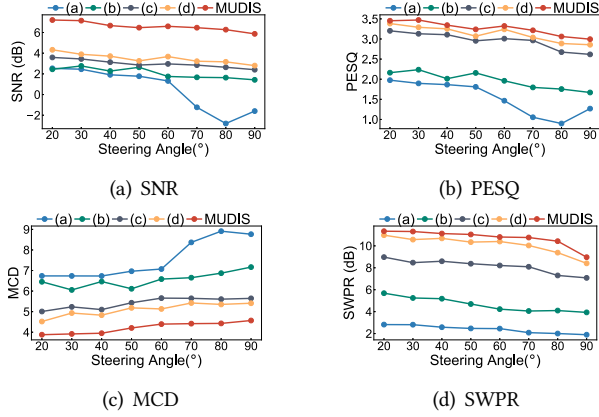


Figure 18: The overall performance of MuDiS regarding SNR, PESQ, MCD, and SWPR. One beam is always targeted 0°, and another steers from 20° to 90°.

4.3 System Performance

Overall Performance. In this section, we evaluated the overall performance of MuDiS. MuDiS projects multi-beam to multi-directions, and each beam plays an English sentence. We test each part of the proposed approaches one by one. For each scheme, we randomly test 30 sets (20 TTS and 10 vocal recording clips). One beam is always delivered to 0°, and another beam is steered from 20° to 90°. Fig. 18 shows the performance when using part of approaches. Results indicate that metasurface can improve steering angle, as a flat curve during whole angle ranges. When steering to over 70°, the performance without metasurface declines sharply due to grating lobes produced by 0°. The distortion reduction scheme mainly improves SNR for effectively reproducing original audio content. Leakage cancellation and phase robustness schemes are significant to speech quality (PESQ, MCD) and leakage suppression (SWPR).

Since no COTS multi-directional speaker can offer all targeting features (independent beams, digital steering, and leakage-suppression), we chose two baselines as a comparison to better clarify the performance of MuDiS: a smartphone speaker and a commercial-of-the-shelf (COTS) directional speaker (LISENECHO G-19). The average results are shown in Tab. 1. The smartphone speaker cannot support directional projection as its SWPR is only 0.86dB. The COTS directional speaker and MuDiS with one beam perform slightly worse than it as they sacrifice the audio quality for directivity. When project only 1 beam, MuDiS performs better than COTS directional speaker regarding SWPR and SNR because the metasurface eliminates the grating lobes and the distortion reduction scheme improves the speech quality. However, the COTS directional speaker cannot support multi-beam, as MuDiS can achieve close performance when projecting 2-beams (for 6.56dB SNR, 3.25 PESQ, 4.31 MCD, 10.97 SWPR).

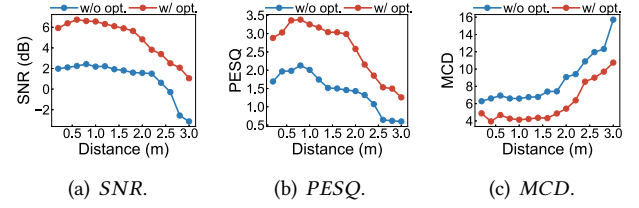


Figure 19: Performance of varying the distances.

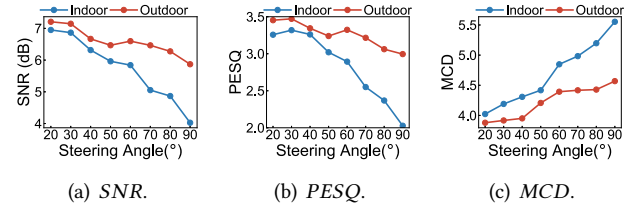


Figure 20: Performance of varying the environments.

Impact of Distance. We evaluated the impact of varying hearing distances and observed the results shown in Fig. 19. At very close distances (i.e., 20cm – 50cm), MuDiS performs slightly worse because the angular differentiation is not apparent and the beam is not fully formed. When the distance is between 0.5m and 2m, MuDiS always performs well. When the distance exceeds 2.5m, PESQ drops significantly below 1.5. Despite this, 2m supports many scenarios, such as homes and museums. For 3-beams and 4-beams, MuDiS can support 1.5m and 1m distance, respectively. To support greater distance, one possible solution is to increase the number of ultrasonic transducers. Moreover, compared with traditional multi-beamforming, MuDiS outperforms in whole distances.

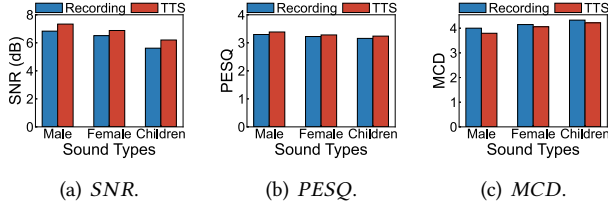
Impact of number of beams. MuDiS also supports projecting 3- and 4-beams with acceptable performance, shown as Tab. 1. The null width is set $\pm 10^\circ$ for a beam. As the number of beams increases, the interference becomes stronger. Despite the interference, MuDiS can still achieve 2.88 PESQ and 2.67 PESQ in 3- and 4-beams projection, respectively.

Impact of Environments. We evaluated MuDiS in two environments: an open-space office and an outdoor one. In the office, which is 6m × 8m large with desks and chairs, MuDiS is placed on top of a desk close to the wall and facing toward the user sitting at the desk. Here is something within 20m of the device in the outdoor environment. From Fig. 20, We can see that in both environments, MuDiS achieves high PESQ scores (> 2.7), demonstrating its efficacy.

Impact of Sound Types. We also evaluated the impact of varying sounds and observed the results shown in Fig. 21. 15 TTS sentences and 15 vocal recording sentences are played using MuDiS, and both include three types: male, female, and children voices. The results show that the voices of males outperform other types because low frequencies are more dominant in male voices. TTS sentences are slightly better than recording voices since they have no noise. Despite this,

Table 1: The performance comparison of MuDiS under different schemes.

		Beam Num.	Methods					Metrics			
			Beam-forming	Meta-surface	Leakage Cancellation	Phase Robust	Distortion Reduction	SNR	PESQ	MCD	SWPR
Different Schemes	(a)	2	✓	✗	✗	✗	✗	1.04	1.66	7.30	2.48
	(b)	2	✓	✗	✗	✗	✗	2.18	2.01	6.59	4.97
	(c)	2	✓	✓	✓	✗	✗	2.72	3.11	5.29	8.24
	(d)	2	✓	✓	✓	✓	✗	2.80	3.16	5.22	10.58
MuDiS	2-beams	2	✓	✓	✓	✓	✓	6.56	3.25	4.31	10.97
	3-beams	3	✓	✓	✓	✓	✓	5.85	2.88	5.26	8.62
	4-beams	4	✓	✓	✓	✓	✓	5.66	2.67	6.03	6.93
	1-beam	1	✓	✓	✗	✗	✓	6.96	3.42	4.14	18.95
COTS Directional Speaker		1	✓	-	-	-	-	6.07	3.40	4.70	15.37
Smartphone Speaker		-	✗	-	-	-	-	8.52	4.02	3.04	0.86

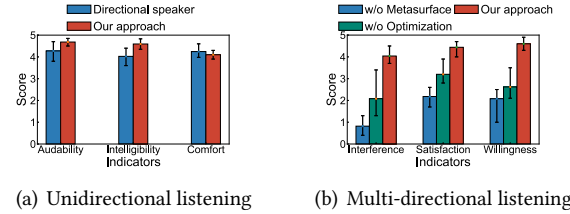
**Figure 21: Performance of varying the types of sounds.** the PESQ still achieves over 3 and MCD below 4 for all types of voices, which verified the ability of MuDiS to play speech.

4.4 User Study

We recruited 10 volunteers (aged between 20 and 55; 5 female and 5 male) to evaluate the system's performance. The study took place in an $18ft \times 24ft$ room, with participants positioned $1.5m$ away from the speaker. They initially heard iPhone 12 audio, then evaluated the directional speaker and MuDiS's performance on playing the same audio. Volunteers were instructed to rate the experience on a scale of 1 to 5, with higher scores signifying superior experience. The user study involves two stages below:

Unidirectional listening. In this part, we mainly compared the speech quality of MuDiS with a COTS directional speaker since there is no COTS multi-directional speaker that offers all targeting features. The evaluation includes three metrics: 1) *Audibility*: whether the sound heard is loud and clear enough without noise; 2) *Intelligibility*: ability to understand meaning from sound; 3) *Comfort*: measures the impact of high-frequency signals on the human ear and whether they feel uncomfortable. Fig. 22(a) shows MuDiS is superior in audibility and intelligibility. However, since high-energy ultrasound still exerts sound pressure on human ears, their comfort ratings are only around 4.

Multi-directional listening. Here, we mainly focused on examining mutual interference, which compared the performance of our proposed MuDiS with schemes without meta-surface and optimization. The participants were divided into two groups located $1.5m$ away from the speaker at different groups of preset angles ($[0^\circ, 30^\circ]$, $[0^\circ, 60^\circ]$, $[-30^\circ, 30^\circ]$). The volunteers are divided into two groups, who stand at different angles to listen to different audio contents and score on three metrics: 1) *Interference*: assess whether the unintended

**Figure 22: The user study scores of unidirectional listening and multi-directional listening.**

audio content can be heard. A higher score indicates lower leakages. 2) *Satisfaction*: rates the comprehensive experience, including practicality, sense of experience, and other potential dimensions. 3) *Willingness*: the desire of volunteers to use MuDiS in life, which includes other factors like cost and size. Fig. 22(b) shows that our scheme outperforms others in all three metrics, as those schemes may hear disturbing leakages from another direction. Our user study indicates that MuDiS performs well as a multi-directional speaker.

5 RELATED WORK

5.1 Multi-Beamforming

RF Multi-Beamforming. Based on the multi-channel digital estimation and transmission in MIMO [36, 72], multi-user MIMO (MU-MIMO) enhances wireless communication capabilities by accommodating multiple users. Space-division multiple access (SDMA) [61], massive MIMO [31] and co-ordinated multipoint (CoMP) [14] exploit spatial degrees of freedom to deliver distinct content to different users. Further extension of optimization [9, 22, 35, 45, 55, 69] in beamforming from analogue phase shifters to multiple RF chains for digital beamforming enables multibeam transmission.

Acoustic beamforming. Acoustic beamforming mainly falls into two categories: 1) *Audible multi-beamforming*: In this range, multi-angle content projection is achieved using large acoustic systems due to the extended wavelengths involved [16, 65, 66, 71, 91]. Chang, et al. [16] and Tang, et al. [71] use linear arrays, but their sound fields vary across frequencies. [65, 66, 91] use bulky circular arrays (ranging from $60cm \times 60cm$ to $150cm \times 150cm$), while MuDiS is relatively compact, measuring $8cm(W) \times 15cm(H)$. 2) *Ultrasonic multi-beamforming*: These applications [12, 62] primarily

employ multi-beamforming for sensing and communication purposes. Furthermore, the utilization of *parametric arrays* to generate low-frequency sounds through nonlinearity finds significant application in underwater acoustic engineering [27, 64, 79] and directional speaking [73, 88]. Those work connect standard ultrasonic transducers, inevitably suffer from grating lobes and limited steering angle ($\pm 40^\circ$) [54]. [63] supports digital steering but all beams play identical audio. COTS multi-directional speakers face similar issues like mechanical steering (Rocketfish [58], Sounddynamics [68]), limited steering angle (SXS [70]), or identical audio content (PAN Acoustics [4–7]) while being expensive (ranging from \$1700(SXS) to \$5500(PAN Acoustics)). In comparison, MuDiS is relatively cost-effective (\$350) which leverages metasurface to ensure proper element spacing and wavefront, enabling wide-angle steering (over 130° HPBW), grating-lobe elimination, and distinct content projection.

5.2 Acoustic Metasurface

Acoustic metasurfaces [10] are artificial structures capable of manipulating the phase and/or amplitude of acoustic waves, including various types [13, 48, 49, 83, 84]. They can provide controllable acoustic transmission for acoustic sensing and communication, enable functionalities like beam steering [47, 80] and focusing [8, 44]. Yang et al. [25] proposed a 3D printed structure covering a microphone, enabling angle-of-arrival detection with a single microphone. Zhang et al. [90] achieved adaptive beam steering with an optimized metasurface. However, this approach entails a fixed metasurface position and correlates metasurface size with speaker array size. Moreover, it doesn't address the issue of grating lobes stemming from increased speaker spacing.

Our metasurface relatively reduces the size to enhance portability. Specifically, our inspiration resembles the metasurface in [11], which covers the speaker to reshape the emitted wave. Importantly, our design accounts for the influence of reflected waves on the sound source. This adaptation of our metasurfaces for large ultrasonic transducers enhances their capacity to achieve superior multi-beamforming performance while effectively mitigating grating lobes.

5.3 Acoustic Nonlinearity

Acoustic nonlinearity [75, 76] can manifest in electronic components such as speakers and microphones [60, 89], as well as in propagation media like air [24] and water [92].

Hardware nonlinearity. Nonlinearity arises within the diaphragm and amplifier of a microphone. Recent studies [18, 43, 59, 60, 89] demonstrate the feasibility of injecting inaudible voice commands by exploiting the hardware nonlinearity. Moreover, microphone nonlinearity can be harnessed to enhance sensing granularity [18].

Media nonlinearity. Researches [74, 77, 82] verified the distortion of sound during its propagation within nonlinear media, including air and water, which suggests ideas for ultrasound demodulation in air. Projects [53, 86] such as SoundLazer [38] proposed highly directional speakers grounded in air nonlinearity. The acoustic nonlinearity of sound transmission underwater has broadened the application in underwater acoustic engineering, such as sub-bottom profile measurement [26, 32], underwater acoustic communication [39, 85], and detection of buried targets [15, 42].

6 DISCUSSION

MuDiS is designed to serve as a multi-directional speaker for many applications. We believe MuDiS has the extensive applications in the future if iteratively upgraded:

Limited bandwidth. Due to the physical limitations of a piezoelectric transducer, the bandwidth of sound waves emitted by transducers is usually below 4kHz , which limits the frequency range of recovered audio. Although it is enough for speech restoration, we hope MuDiS can further support the music. Combining multiple transducers of various central frequencies is possible to achieve a wider bandwidth.

Multipath and occlusion. In more crowded indoor environments (e.g., in a car), the multipath effect that reflects sound waves may generate more interference. Channel estimation may be helpful to eliminate this issue. In addition, the obstacles will block the transmission of sound waves. It is possible to bypass the obstacles by leveraging multipath. We leave them in our future work.

Safety concern. In this work, we implemented MuDiS using transducers with central frequency of 21kHz . The transmission power complies with FDA safety (80dB SPL for 21kHz at 1m). Current prototype can support close-up scenarios like museums. Considering the safety concerns and long distance demand, we will switch to a higher frequency and design accordingly to support longer distance in the future.

7 CONCLUSION

In this paper, we present MuDiS, a novel multi-directional speaker using a parametric array. We designed a metasurface to overcome the physical limitations of an ultrasonic transducer and proposed an optimization scheme to enhance speech quality. Extensive experiments verified the capability of MuDiS to realize multi-directional audio projection without leakages.

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